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Underwater Structure Inspection and Analysis by 3D Subsea Photogrammetry Technology

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Abstract

Petroleum Nasional Berhad (PETRONAS) engineered, fabricated, and installed offshore wellhead light weight structure known as Platform A, at 70m water depth with bridge link to the existing platform, offshore Peninsular Malaysia. During offshore installation of the substructure, the top section of main leg was deflected laterally by approximately 2.65 m from the target position. Post inspection survey by remotely operated vehicle (ROV) during offshore installation time has confirmed there was a deflection on the substructure however the survey method has been known to have lower accuracy while substructure structural integrity assessment required precise input data for valid results. Further analysis to verify the substructure structural integrity is done by performing In-Place and Fatigue analysis considering residual stresses based on actual as-installed condition of the substructure. As part of the assessment, 3D subsea photogrammetry survey technology has been deployed to verify the substructure verticality and deflection using accurate underwater survey measurement technique. The emerging 3D subsea photogrammetry survey technology was used to achieve high accuracy and stringent subsea relative survey measurement, replacing the conventional method using diver and less accurate underwater positioning system by Ultra-short baseline (USBL) which is not comprehensively covered the entire substructure components. The method avoided the complex survey and positioning systems set-up that would consume longer installation and calibration time and pose greater Health, Safety, Security and Environment (HSSE) risks and exposures during operation. The 3D subsea photogrammetry survey was executed by utilizing 3D imaging video camera attached on the work class ROV which integrated with the survey-grade subsea gyrocompass, motion sensor and interfaced with navigation system to achieve accurate and seamless underwater photogrammetry survey. 3D model of the substructure was generated for accurate 3D model analysis and study for structural integrity verification. The reliable survey data has allowed structural integrity reassessment to be conducted to finalize the analysis report through modeling refinement, content presentation to reflect latest 3D survey report and would not jeopardize the current fitness-of-purpose structural integrity of Platform A. The results demonstrated that the 3D subsea photogrammetry survey achieved centimeter level accuracy in relative surveying measurement, while gaining operational cost benefits by diverless operation, reducing

operational time where the survey was conducted simultaneously with General Visual Inspection (GVI) by the ROV and eliminating uncertainty on data accuracy and coverage. It also reduced HSSE risks exposure by minimizing ROV diving and tasking operations and improved situational awareness by real-time video and continuous high definition (HD) photo taking to generate photo realistic model of point cloud. This paper discusses the implementation of the 3D subsea photogrammetry survey technology, highlighting its technical capabilities, challenges encountered during offshore operations, perceived value creation, and the valuable technological experiences gained. These insights can be leveraged for broader subsea inspection, installation and monitoring applications.

Introduction

As the oil and gas industry continues to seek cost-effective and modular solutions for developing marginal fields, innovative structural designs and installation methodologies are being explored to optimize offshore asset design. One such innovation is the Platform A light weight structure, an engineered tripod-based wellhead platform designed for shallow to moderate water depths and minimal topside weight. This novel substructure design features an unconventional tripod-based substructure that introduces separated components to enable utilisation of smaller lifting capacity crane leading to a more cost-effective heavy lift vessel for installation.

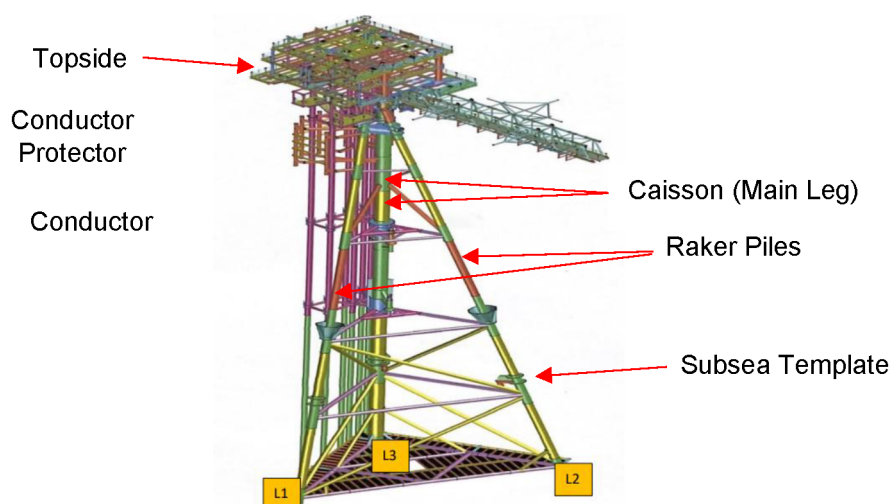


Figure 1—Platform A Perspective View

During offshore installation of Platform A, while driving the raker pile at Leg L2, the top of caisson deflected laterally by approximately 12.8m from the target position. The deflection was improved from 12.8m to 2.65 m after jacking activities. Further analysis to verify Platform A structural integrity is done by performing in-place and fatigue analysis based on actual as installed condition of substructure. After substructure levelling completed, it is expected that the residual stress will be generated (especially at caisson and raker pile) which needed to be verified. Based on engineering re-analysis incorporating the deflected conditions, it is concluded that the as-installed Platform A structural integrity satisfy design requirement. However, as part of way forward and recommendation, General Visual Inspection (GVI) by Remotely Operated Vehicle (ROV) to be performed during the baseline survey campaign. Since reliable survey data is crucial to assess the structural integrity of the substructure section, 3D photogrammetry survey technology was mobilized to inspect the suspected deflection area at leg L3 and structure verticality assessment during baseline survey works. The primary objective of the underwater baseline survey campaign is to conduct a comprehensive baseline inspection of the newly installed structures. This survey

is crucial in establishing reference data on the condition and configuration of the platform, serving as a benchmark for future integrity risk-based inspections and maintenance planning.

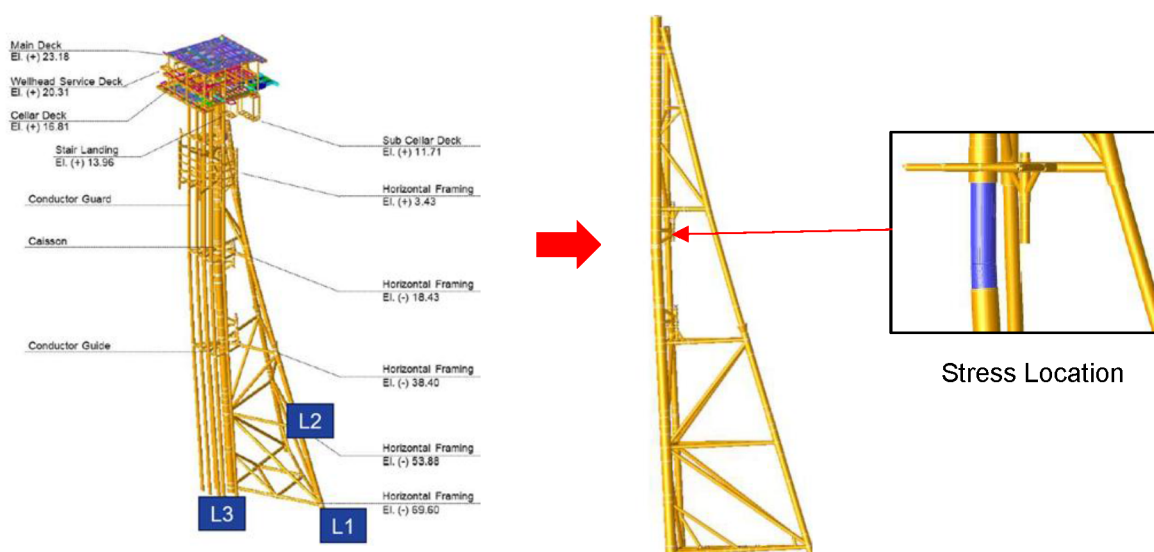


Figure 2—Critical Regions for Installation Residual Stress

Case For Change

During offshore installation of Platform A, post installation ROV inspection detected a deviation of the substructure from its intended position. Initial assessment using USBL based methods indicated lateral deflection, however, due to limitations in precision and resolution, further analysis required more accurate spatial data ensuring the structural integrity of offshore installations is critical for the safe and efficient operation of oil and gas facilities. Traditional subsea inspection techniques, such as diver-assisted visual inspections or positioning using acoustic Ultra-Short Baseline (USBL) systems, often lack the precision required for detailed structural assessments. These methods may also introduce operational risks, longer survey durations, and data reliability concerns, particularly in complex subsea environments. In response to these challenges, PETRONAS implemented 3D subsea photogrammetry survey technology which is more accurate and high resolution during the offshore baseline survey campaign of Platform A to reassess the as-installed condition and support structural integrity verification through In-Place and Fatigue analysis. To address this, 3D subsea photogrammetry was adopted as a primary method for generating a high-resolution digital 3D model of the substructure. The objective is as follows:

- a. Accurately determine substructure verticality and deflection
- b. Provide reliable data for structural analysis
- c. Reduce operational risks, HSSE issue and downtime

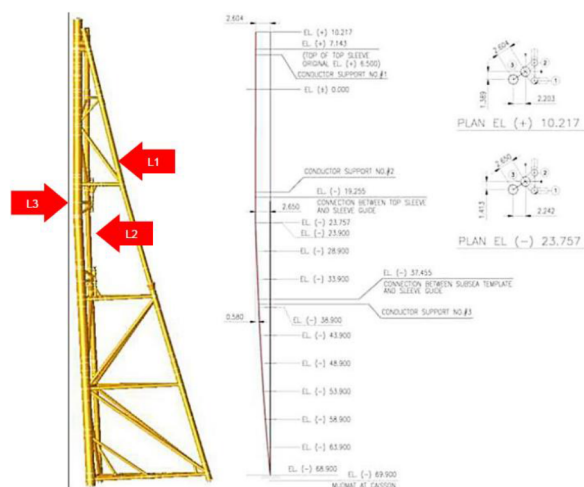


Figure 3—Caisson Leg L3 Elevation Profile Measured by ROV Using Ultra Short Baseline (USBL) Positioning System

Survey Equipment and System Configuration

Subsea photogrammetry survey utilizes a series of overlapping images captured underwater to reconstruct accurate 3D models. Specialized cameras and imaging systems from MCS Photo Realistic 3D Cloud (PRC) was mounted on Remotely Operated Vehicles (ROV) and capture multiple images from different perspectives. These images are then processed using photogrammetry software to create a dense point cloud and generate a 3D model of the surveyed area. The 3D PRC subsea photogrammetry survey was executed by utilizing 3D imaging stereo cameras. This system is self-dependent and doesn't require integration with any ROV positioning sensors thus the positioning accuracy does not affect the model accuracy for relative measurement.

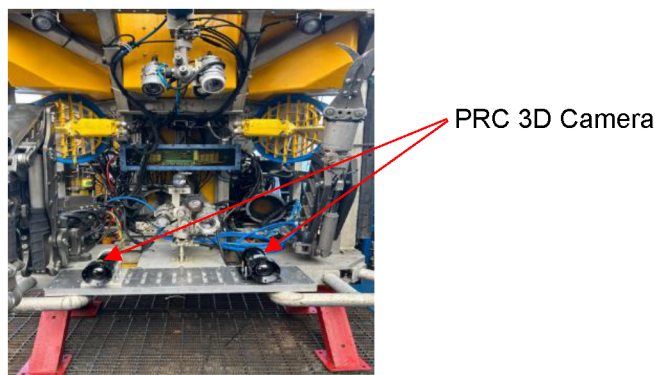


Figure 4—Camera Configuration Mounted on Work Class ROV

Table 1—3D Camera Specification

3D Camera Specification	
Resolution	2750 (H) × 2200 (V); 6.1 MP
Frame Rate	6 fps
Power Consumption	5.4W at 12VDC
Camera Dimension	92 × 53.3 × 33 mm
Mass	224 g

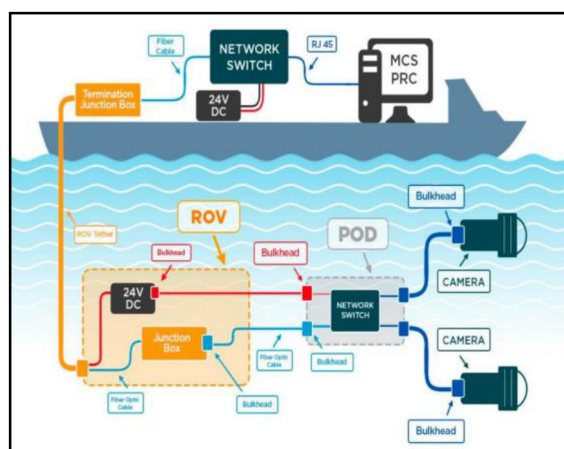


Figure 5—Network Diagram for PRC System

Data Acquisition

ROV operations were synchronized with General Visual Inspection (GVI) runs to maximise operation efficiency. A predefined ROVs path and imaging plan ensured complete coverage of all critical structural components. Redundant imaging from multiple perspectives reduces occlusions and enhanced model accuracy. Acquisition software used to capture and record 3D photo during online survey was MCS PRC Camera Controller V3.0. 3D PRC acquired the HD photos with rate 6 HD Images frame per second and 60 to 80% image overlap. Data served as inputs for Finite Element Analysis (FEA) to support structural In-Place and Fatigue life evaluations.

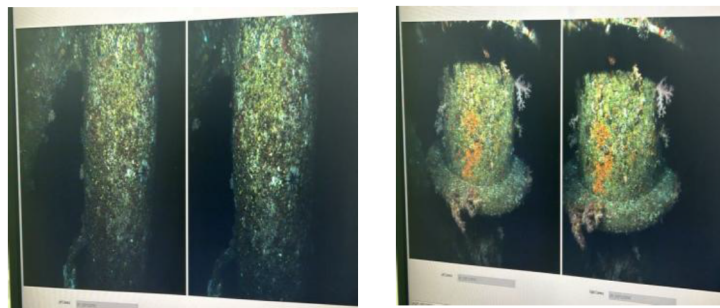


Figure 6—Image at Water Depth -20 m (left) and Water Depth -50 m (right)

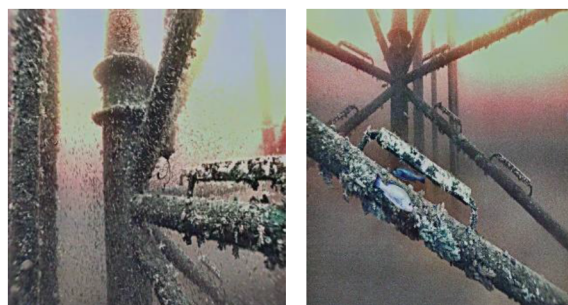


Figure 7—Image at Bracing Area Data Processing and Quality Check

Photogrammetric data were processed offshore and refined post processing onshore using 3D reconstruction algorithms. The generated 3D model was compared with the engineering reference model to quantify deflection, inclination, and structural alignment. The MCS 3D PRC software package used for data processing phase is as per below:

- a. MCS Divide Images: Used to select and identify the images and frames to be used as the input for 3D cloud model generation.
- b. MCS Image Enhancer: The software is used to enhance the images quality in poor visibility areas
- c. MCS 3D Reconstruction: The processing software where the data acquisition and calibration factors are inserted along with the selected images to produce 3D cloud model in .ply universal format, further noise filtering could be applied on the output model.
- d. MCS 3D Cloud Viewer: .ply viewer software with measuring capabilities to view the processed model of the underwater asset
- e. AutoCAD: The MCS 3D PRC model could be further exported to AutoCAD to build 3D Solid model of the structure and perform further metrology measurements.

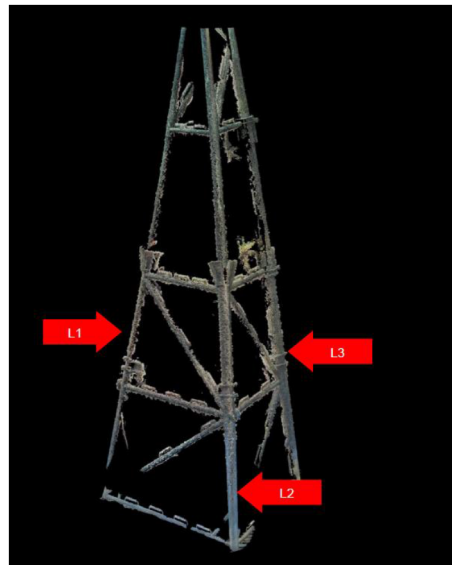


Figure 8—Substructure Point Cloud Data (Generated from image processing)

Normally the cameras arrive to the working site was pre-calibrated from the workshop. The camera calibration was performed to ensure optimal functioning and production of reliable image data, minimizing potential errors from the outset. A meticulous triple check system was conducted to evaluate crucial aspects of the process from various perspectives as per following:

Table 2—Camera Calibration

No.	Calibration	
1	Camera parameter validation	Verify intrinsic parameters (focal length, principal point, radial and tangential distortions).
2	Distortion analysis	- Employ checkerboard or grid patterns for calibrating distortion - Calculate lens distortions (barrel, pincushion) and apply corrective algorithms.
3	Environmental adjustments	Factor in environmental conditions such as temperature, pressure, and humidity, which can affect lens curvature and response.

Various Quality metrics are applied to ensure effective and accurate measurements and calibrations such as evaluating Mean Squared Projection Error (MSRE) which validate the accuracy of image alignment by

comparing the expected projection of a point in 3D space onto an image plane with its actual measured location, acceptable Mean Squared Projection error (MSRE) should be less than camera pixel size.

The Mean Squared Reprojection Error (MSRE) evaluates the accuracy of image alignment by comparing the expected projection of a point in 3D space onto an image plane with its actual measured location. It's calculated as:

$$MSRE = \frac{1}{n} \sum_{i=1}^n (e_i^2)$$

Where e_i is the error for each projected point.

- After aligning images, the software calculates reprojection errors for tie points in the software's result files "reprojection error metrics".
- Acceptable Range: MSRE should be below the pixel size of the camera (<0.2 pixel for precise photogrammetry).

Standard deviation (SD) measures the consistency of errors. It identifies how much variation exists from the mean value of reprojection errors.

$$SD = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

A low SD indicates that errors are uniformly distributed across the dataset. A high SD may reveal issues such as misaligned images or outliers in the data.

Table 3—Dimensional Difference

Point Number	Elevation (-m)	Reference Design Dimension	PRC Value	Deviation
1	64.90	1.000	1.12	0.120
2	65.90	0.964	1.05	0.086
3	66.90	1.020	1.07	0.050
4	67.90	1.098	1.11	0.012
5	68.90	1.080	1.08	0.000
Standard Deviation				0.050

Interpreting Results:

- SD < 0.4 pixels: Excellent alignment and reconstruction accuracy.
- SD between 0.4–1 pixel: Acceptable but may require minor adjustments.
- SD > 1 pixel: Indicates potential errors requiring immediate correction.

Interpretation of Results (MSRE and SD)

- Low MSRE + Low SD: Excellent alignment and reconstruction quality.
- High MSRE + Low SD: Consistent but offset errors; calibration or control points might need review.
- High MSRE + High SD: Indicates systemic errors (e.g., calibration issues).

Model Dimensions Validation (Scale and Length)

- Dimension Extraction: Extract 3D model dimensions (e.g., caisson length, pipe diameter) from PRC point cloud using MCS 3D Viewer tools.
- Scale Constraints: Apply constraints using real-world scale measurements or stereo vision cameras to ensure dimensional accuracy.
- Dimension Comparison: Compare measured dimensions with CAD/reference drawings to compute deviations.

Results Analysis and Discussion

The 3D model reconstruction was modelled and overlayed with the coloured point cloud data to allow thorough analysis of the substructure. The 3D photogrammetry survey successfully delivered centimeter level accuracy in measuring the relative lateral deflection and vertical misalignment of the Platform A substructure. Model accuracy and point cloud data were checked with components on the substructure such as anode and bracing.



Figure 9—3D Model overlayed with point cloud data

Overlaying the 3D model with coloured point cloud data enhances visual clarity and enables intuitive understanding of complex geometries. This fusion of geometric and visual data allows analysts and engineers to identify anomalies, deformation patterns, or areas of concern more efficiently than with traditional methods.

Key survey results include:

- Measured tilt and vertical angular deviation of leg L3 is 0.56° with maximum lateral vertical line deflection of 1.425° m at El. (-) 3.17 m.
- Two (2) deflection locations were identified at elevations -37.8 m and -19.8 m
- Full 3D model coverage with photo-realistic point cloud visualisation

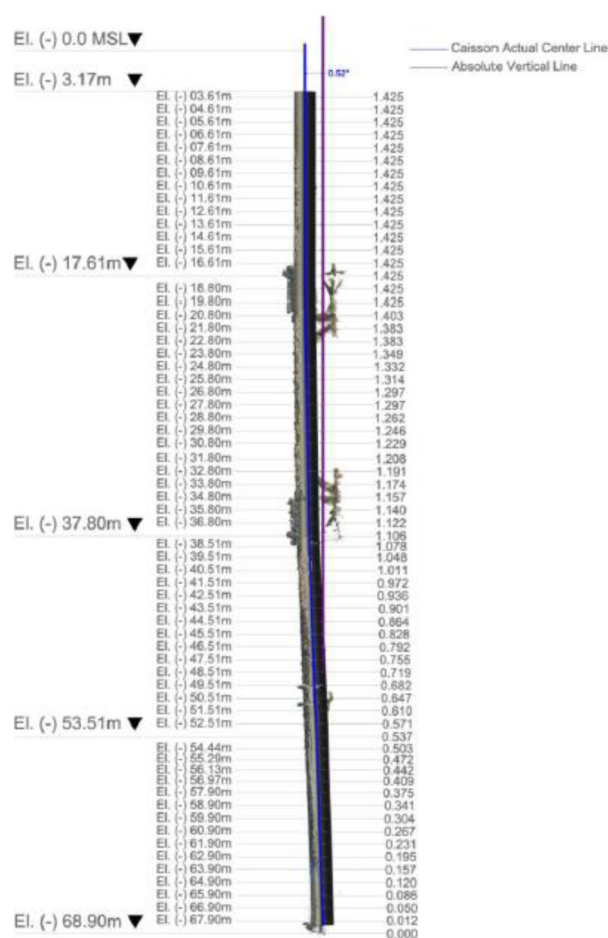


Figure 10—L3 verticality and Lateral Deflection Analysis

The 3D photogrammetry survey achieved centimeter-level accuracy, which is crucial for evaluating structural integrity in subsea environments. This level of precision allows for reliable measurements of lateral deflection and vertical misalignment, which are key indicators of structural deformation in offshore platforms like Platform A.

The caisson lateral deflection profile from the platform baseline survey shows a smaller lateral deflection profile compared to the original as-built.

Survey analysis is also performed at other critical structural members such as mudmat to ensure thorough assessment for structural integrity check. It was measured that:

- Slope angle value between L2 – L3 = 0.74°
- Slope angle value between L1 – L3 = 1.05°

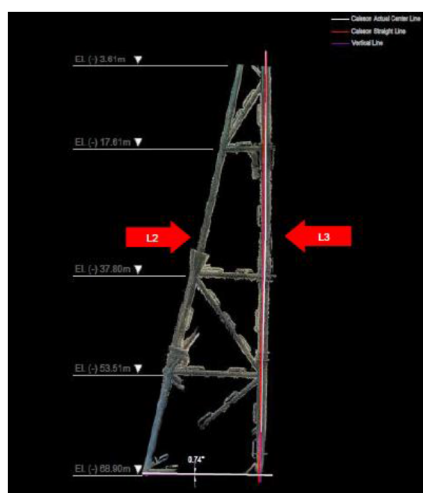


Figure 11—Mudmat Inclination Angle L2 - L3 Side



Figure 12—Mudmat Inclination Angle L1 - L3 Side

By cross-referencing the model and point cloud data with known components on the substructure (such as anodes, mudmat, and bracing), the analysis validates the accuracy and reliability of the photogrammetric outputs. This step is critical for establishing confidence in the digital model for decision-making and future monitoring.

The mudmat inclination angle from the platform baseline survey shows a smaller inclination compared to the original as-built.

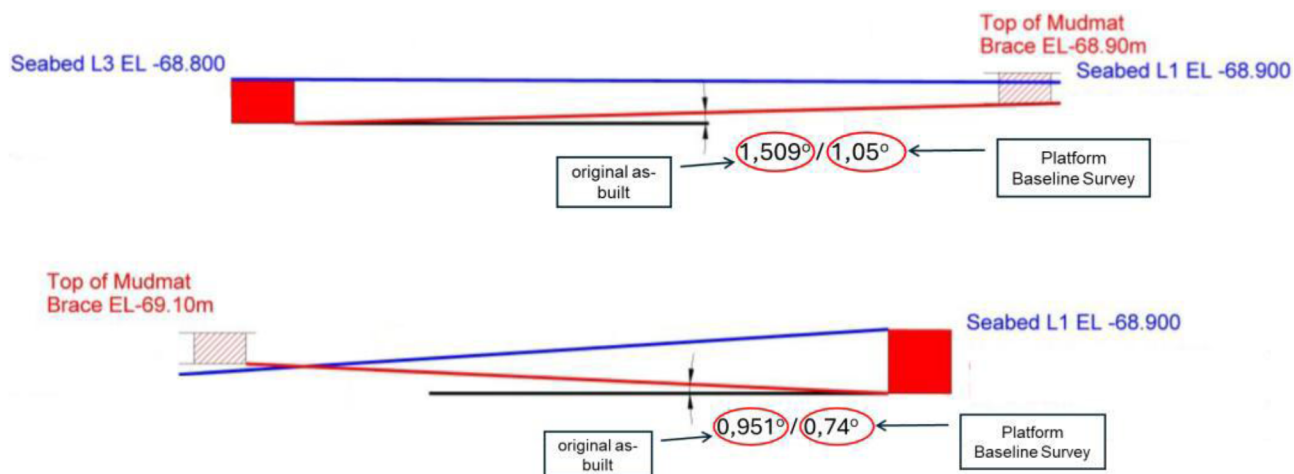


Figure 13—Mudmat Inclination Profile Comparison Between Original As-Built and Baseline Survey

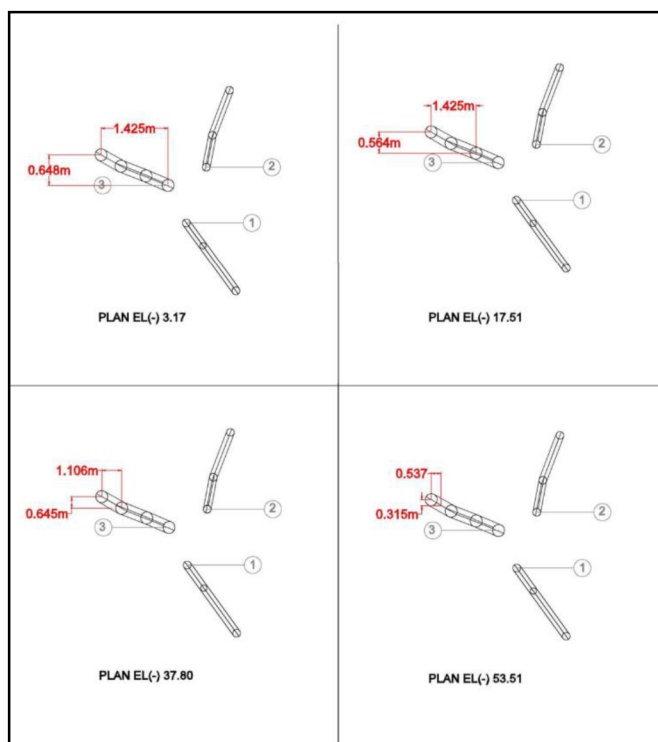


Figure 14—Platform A Substructure Lateral Deflection – Plan View

The Final results and analysis at area of interest were summarised as follows:

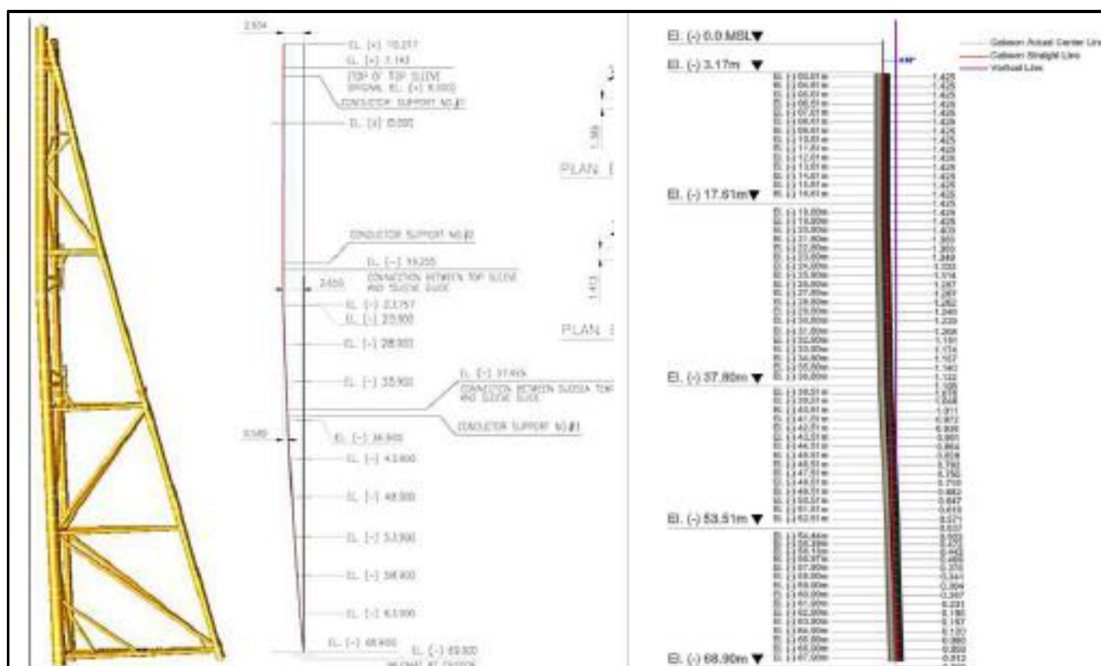


Figure 15—Survey Results Comparison Between USBL and Photogrammetry

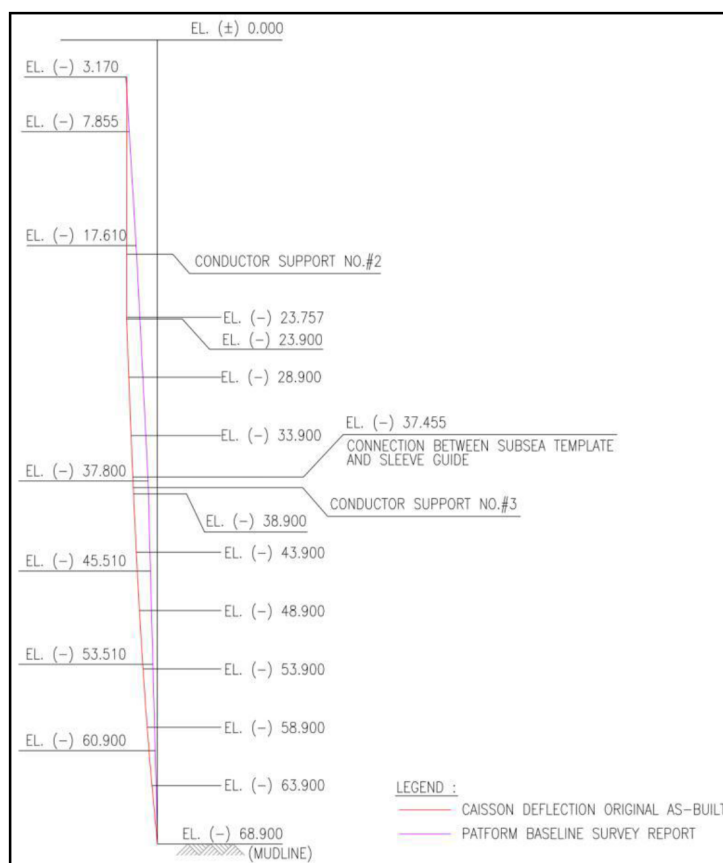


Figure 16—Caisson Lateral Deflection Profile Comparison Between Original As-Built (USBL) and Baseline Survey (Photogrammetry)

3D subsea photogrammetry offers significant advantages over USBL for underwater structural measurement and inspection tasks. While USBL is primarily used for tracking the position of underwater

vehicles or equipment, providing meter-level positional accuracy, subsea photogrammetry delivers centimeter-level precision along with high-resolution 3D models and visual detail. This makes subsea photogrammetry ideal for detecting small deformations, misalignments, or deflection on subsea structure of Platform A.

Table 4—Deflection Angle Comparison

Elevation (m)	Measurement by USBL	Measurement by Photogrammetry	Face
−40.50	0.580	1.011	L1-L3
−21.80	2.650	1.383	L1-L3
−3.61	2.604	1.425	L1-L3

Table 5—Inclination Angle Comparison

Face	Measurement by USBL	PRC Results (Degree)
L2-L3	0.951	0.74
L1-L3	1.509	1.05

The latest 3D subsea photogrammetry survey results were using precise subsea survey measurements method compared to the previous measurement method by USBL. The results verified the previous measurement data and shown reliable deflection values as compared to the previous as-installed condition. Previous Structural In-Place Re-analysis was performed based on initial as-installed survey data which adopted higher and more conservative deflection values. The latest structural integrity is found to be adequate and satisfactorily meeting code requirements.

Compared to conventional USBL or diver-based surveys, the photogrammetry method offered:

- Improved spatial resolution and measurement accuracy
- Reduced survey duration and ROV tasking time
- Enhanced safety by minimizing the need for complex sensor setups or diver involvement

Furthermore, real-time visualization enhanced situational awareness during operations, allowing for prompt decision-making and data validation offshore. **3D Subsea Photogrammetry Technology Advantages and Lesson Learned**

Advantages:

- High accuracy in relative positioning and geometry reconstruction.
- Non-intrusive and diverless operation for faster data acquisition.
- Self-dependent and doesn't rely on other sensors for data acquisition, processing and reporting
- No site calibrations are required
- Auto-Scaling, no reference objects are required underwater
- Lower HSSE exposure.
- Effective execution with concurrent survey tasks.

Lesson Learned:

- Lighting and turbidity conditions must be managed for optimal image clarity. ROVs should be equipped with good lighting system to enhance image quality.
- ROV flight path planning is critical to prevent occlusions.
- Sufficient image overlap and coverage are crucial to ensure accurate 3D model reconstruction. Proper planning and maneuvering of the imaging system or vehicle are necessary to capture the required image data.
- Processing large amounts of image data and generating high-resolution 3D models can be computationally intensive and time-consuming, requiring powerful hardware and specialized software.
- Marine growth on substructure might influence the accuracy of the 3D model reconstruction. Proper modelling techniques should be practiced with reference to engineering drawings.
- Early integration with structural engineering teams and ROVs team ensures seamless operation and use of data for structural modelling analysis.

Recommendations for Scaling Up Underwater Photogrammetry Survey Technology

To scale up the use of underwater photogrammetry for other exploration and development activities, the technology should be applied across all offshore assets, from jackets and risers to subsea manifolds and pipelines, and integrated into both new projects and existing asset integrity and inspection program. Photogrammetry outputs can be linked with digital twin platforms for real-time condition monitoring, predictive maintenance, and accurate as-built records. Surveys should be combined with other subsea tasks during a single deployment to reduce vessel time and HSSE exposure, supported by clear operating procedure, training, and a central data repository to ensure quality and consistency. Piloting the method in frontier and deepwater locations using ROVs or other deployment platforms will demonstrate its adaptability, while cost and time savings, along with HSSE benefits, should be communicated to stakeholders to secure wider adoption. Collaboration with technology providers can further enhance capability through real-time processing, AI-driven analysis, and integration with other subsea sensors, making subsea photogrammetry a standard and efficient tool for offshore inspection and development project.

Conclusion

3D subsea photogrammetry survey technology was successfully implemented and executed for Platform A structural integrity and baseline survey scopes. This approach eliminated the need to deploy a separate inspection-class ROV, resulting in a 30% reduction in operational costs. The strategized operation has optimized the survey timeline, reducing the survey operation cost and time from 12 hours to 4 hours, minimizes HSSE risk exposure during offshore execution. The deployment of 3D subsea photogrammetry technology in the inspection and analysis of Platform A demonstrated a significant advancement in underwater inspection and metrology. The technology provided a precise and cost-effective solution for post-installation, support low carbon operation, improve integrity assessment and baseline survey quality, replacing traditional methods that lacked the necessary accuracy and operational efficiency. The success of this survey highlights the growing value of photogrammetry in subsea applications and supports its adoption as a standard tool for structural inspection, installation verification, and digital twin modelling in offshore oil and gas operations.

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